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Lec # 17

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Graph $\rightarrow$ functionCurve $\rightarrow$ EquationLine Integrals "Curve Integrals"

In this section, we define an Integral that is similar

Def:- Let F be a continuous vector field defined on a smooth curve C given by a vector function $\gamma(t)$ $a \leq t \leq b$. Then the line integral of F along C is,

$$\boxed{\int F \cdot d\gamma = \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt} \quad \gamma(t) = x\mathbf{i} + y\mathbf{j}$$

$$F(x, y, z) = F(x(t), y(t), z(t)) = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$$

$$d\gamma = \gamma'(t) dt$$

$$\gamma(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$$

Exp:- Find the work done by the force field

$F(x, y) = x^2\mathbf{i} - xy\mathbf{j}$ in moving a particle along the quarter circle $\gamma(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$, $0 \leq t \leq \pi/2$.

Sol:-

$$F(x, y) = x^2\mathbf{i} - xy\mathbf{j}$$

$$\gamma(t) = (x(t), y(t)) = \cos t\mathbf{i} + \sin t\mathbf{j}$$

$$x = \cos t, \quad y = \sin t$$

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$$\int \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} F(x(t), y(t)) \cdot r'(t) dt$$

$$= \int_0^{\pi/2} (\cos^2 t \mathbf{i} - \cos t \sin t \mathbf{j}) (-\sin t \mathbf{i} + \cos t \mathbf{j}) dt$$

$\mathbf{i} \cdot \mathbf{i} = 1 \quad , \quad \mathbf{j} \cdot \mathbf{j} = 1$

$$= \int_0^{\pi/2} (-\cos^2 t \sin t - \cos^2 t \sin t) dt$$

$$= \int_0^{\pi/2} (-2 \cos^2 t \sin t) dt = 2 \int_0^{\pi/2} (\cos t)^2 (-\sin t) dt$$

function's derivative available
Then simple integrate

$$\therefore \int_a^b [f(x)]^n f'(x) dx = \left[\frac{f(x)^{n+1}}{n+1} \right]_a^b$$

$$= \frac{2}{3} \left[\cos t \right]^3 \Big|_0^{\pi/2}$$

$$= 2 \left[\left(\frac{\cos \frac{\pi}{2}}{3} \right)^3 - \left(\frac{\cos 0}{3} \right)^3 \right]$$

$$= 2 \left[0 - \frac{1}{3} \right] = \boxed{\frac{-2}{3}}$$

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Day: MTWTFS

Q. Evaluate $\int_C F \cdot d\mathbf{r}$, where $F(x, y, z) = xyi + yzj + zxk$, and C is twisted cubic given by $x = t$, $y = t^2$, $z = t^3$, $0 \leq t \leq 1$.

Solution:-

$$\mathbf{r}(t) = (x(t), y(t), z(t)) =$$

$$\mathbf{r}(t) = (x(t) + y(t) + z(t)) =$$

$$\mathbf{r}(t) = ti + t^2j + t^3k$$

$$\mathbf{r}'(t) = i + 2tj + 3t^2k$$

$$F = xyi + yzj + zxk$$

$$F(\mathbf{r}(t)) = (t)(t^2)i + (t^2)(t^3)j + (t^3)(t)k$$

$$= t^3i + t^5j + t^4k$$

$$\int_C F \cdot d\mathbf{r} = \int_0^1 F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

$$= \int_0^1 (t^3i + t^5j + t^4k) \cdot (i + 2tj + 3t^2k) dt$$

$$= \int_0^1 t^3 + 2t^6 + 3t^6 dt$$

$$= \int_0^1 t^3 + 5t^6 dt$$

$$= \left[\frac{t^4}{4} + \frac{5t^7}{7} \right]_0^1$$

$$= \frac{1}{4} + \frac{5(1)^7}{7} = \frac{1}{4} + \frac{5}{7}$$

$$= \frac{7}{28} + \frac{20}{28} = \frac{27}{28}$$

$$\begin{array}{r} 2 \overline{) 7.4} \\ \underline{2} \\ 5 \\ \underline{4} \\ 1 \end{array}$$

$$\sqrt{28} \quad 7(1) = 7$$

$$4 \overline{) 28}$$

$$\underline{28}$$

$$4(5) = 20$$

$$7 \overline{) 28}$$

$$\underline{28}$$